Mitochondrion 18 (2014) 12–17

Contents lists available at ScienceDirect

Mitochondrion

journal homepage: www.elsevier.com/locate/mito

Short communication

Mutation-free baby born from a mitochondrial encephalopathy, lactic acidosis and stroke-like syndrome carrier after blastocyst trophectoderm preimplantation genetic diagnosis

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ARTICLE INFO

Article history:
Received 12 April 2014
Received in revised form 18 August 2014
Accepted 18 August 2014
Available online 23 August 2014

Keywords:
Mitochondrial DNA (mtDNA)
Heteroplasmy
Preimplantation genetic diagnosis (PGD)
Blastocyst
Trophectoderm biopsy

ABSTRACT

To investigate the applicability of preimplantation genetic diagnosis (PGD), we used trophectoderm (TE) biopsy to determine the mutation load in a 35-year-old female with mitochondrial encephalopathy, lactic acidosis and stroke-like syndrome (MELAS). Transfer of a mutation-free blastocyst gave birth to a healthy boy with undetectable mutation in any of the analyzed tissues. We found strong correlation among TE cells ($r = 0.90$) within blastocysts and also between cytoplasmic fragments and TE ($r = 0.95$). This is the first case of mutation-free baby born from a MELAS patient after TE biopsy and supports the applicability of blastocyst PGD for patients with mtDNA disorders to establish healthy offspring.

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1. Introduction

Mitochondrial inheritance in mammals is uniparental since mitochondria in the offspring are exclusively transmitted through the maternal lineage (Birky, 2001; Chiaratti et al., 2011; Poulton et al., 2010). More than 150 mitochondrial mutations are reported associated with disorders in human (Levinger et al., 2004). Pathogenic mtDNA mutations usually occur in a state called "heteroplasmy" when cells contain both wild type and mutated mtDNA copies. The proportion of mutated copies is estimated by measurement of the mutation load in the cells. The presence of pathogenic mtDNA does not immediately lead to a fatal outcome of the cells or tissues as wild-type mtDNA copies can partially compensate for the mutated ones. Still, largely depending on the threshold of mutation load and the type of mutation, mtDNA mutations in humans can be associated with severe disorders for which no treatment is currently available. Hence, transmission to progeny should be prevented by developing accurate diagnostic tests or alternative treatment methods, which has generated a great deal of interest in the field of assisted reproductive technology (Poulton et al., 2010).

According to the theory of random segregation of mtDNA during cell divisions and the genetic bottleneck, a stage during oogenesis is seen during which the number of mtDNA copies within the developing precursor oocyte is highly reduced. Hence, in patients who are carriers of mtDNA mutations, some oocytes can be found to be mutation-free. This was shown in earlier studies where whole oocytes were used to determine the mutation load, rendering them not useful for further clinical applications (Blok et al., 1997; Brown et al., 2001). Therefore, preimplantation genetic diagnosis (PGD) could be used to select mutation-free or oocytes/embryos with lowest mutation by determining the mutation load either in polar bodies (PB) or in blastomeres. Importantly, the measured mutation load should be representative for the entire oocyte or embryo. Using heteroplasmic mitochondrial mouse models, levels of heteroplasmy in the first and the second polar bodies correlated well with that found in the respective oocytes and zygotes (Dean et al., 2003; Sato et al., 2005). In contrast, a modest to poor correlation and a wider variability in the levels of mtDNA heteroplasmy were observed in a recent study between PBs and their corresponding oocytes and zygotes in a heteroplasmic mitochondrial mouse model (Neupane et al., 2014). The latter study also showed higher

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http://dx.doi.org/10.1016/j.mito.2014.08.005
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heteroplasmic mtDNA dispersion between the individual blastomeres of different cleavage stage embryos, while a close correlation in the level of heteroplasmy was demonstrated between biopsied trophectoderm (TE) cells and their corresponding blastocysts. TE biopsy could offer a better solution to determine the mutation load more accurately since multiple cells can be biopsied by this method and it can reduce the risk of possible mosaicism. Treff et al. reported that the mutation load determined in TE cells was highly predictive of the inner cell mass cells in human blastocysts (Treff et al., 2012). A successful delivery possessing varying mutation loads (ranging from undetectable to 15%) in different tissues was presented in their study after transfer of an embryo predicted to contain 12% mutation load in a MELAS carrier patient.

In the present study, we performed TE biopsy in a MELAS patient (carrier mother) possessing 16% mutation load in her peripheral blood cells. We verified if TE cells and/or cellular fragments were representative for the whole blastocyst. MELAS is a severe mtDNA disorder mainly caused by an m.3243A>G point mutation in the tRNA (Leu (UUR)) of a mitochondrial genome. The severity of the symptoms is highly variable along with the variation of mutation load in different tissues. We report the birth of a mutation-free baby after transfer of a mutation-free blastocyst determined by TE biopsy. The baby was confirmed to be MELAS mutation-free in three different tissues after birth. To our knowledge, this is the first report of a mutation-free baby born after TE biopsy and PGD analysis. This study will help in preventing transmission of heritable mtDNA disease in patients with pathologic mtDNA mutations to reproduce healthy offspring.

2. Materials and methods

2.1. Patient characteristics and informed consent

A 35 year old female, who was a carrier of the m.3243A>G point mutation (MELAS mutation), visited our department with a profound child wish. A mutation load of 16% was detected in her peripheral blood cells. The patient undersigned a specific informed consent approved by the the local ethical committee of the Ghent University Hospital (EC2009/611) and the Belgian Federal Ethical Committee (Adv037).

2.2. Intracytoplasmic sperm injection (ICSI) treatment and TE biopsy

For controlled ovarian hyperstimulation, the gonadotropin releasing hormone (GnRH) agonist triptorelin (Decapeptyl, Ipsen, 0.1 mg daily for 7 days) and human menopausal gonadotropin (hMG) menotropin (Menopur, Ferring, 300 IU daily for 13 days) were administered. Ovulation was triggered with hCG (Pregnyl, MSD, 5000 IU). The retrieved oocyte–cumulus complexes were denuded with 80 IU/mL hyaluronidase (Cook Medical) by mechanical assistance, and the metaphase II (MII) oocytes were subjected to ICSI with the partner's sperm which showed the following characteristics: 4.7 ml; 10.10^6/ml, 0/79/4/17 (grades of the motility of sperm represented as A/B/C/D, where A + B represents the sum of the percentage of motile sperm, C represents moving sperm, but not progressive and D represents immotile sperm) (World Health Organization, 1999) and 4% normal morphology. Successful fertilization was evaluated 17 h after ICSI by the presence of two pronuclei (PN) and a second PB. Subsequent culture was performed in the sequential culture system Cook Cleavage/Cook Blastocyst media in 6% CO_2 and 5% O_2 until day 6 with concomitant daily scoring of embryos. At day 3 pre-implantation development, a hole was made in the zona pellucida in half of the embryos by applying several laser pulses (Zilos-tk, Alere, Tilburg, The Netherlands) in order to promote herniation of the TE as it forms. In the other half of the embryos, a similar hole was made on day 5 in the morning such that developing inner cell mass (ICM) be located at the 9 o'clock position, fixed to the holding pipette. These blastocysts were incubated for an additional 4 h to allow further blastocoel expansion and spontaneous herniation of the developing TE cells. Blastocyst scoring and grading was performed according to Gardner scoring system (Gardner and Schoolcraft, 1999). Biopsy of herniating TE cells or non-hatching full blastocysts was performed either on day 5 (afternoon) or day in 6. For this, blastocysts were put in Hepes buffered Gamete Buffer medium (Cook). Using a 25 μm biopsy pipette (Cook), gentle suction was applied to dissect on average 4 TE cells by stretching them away from the blastocyst cavity wall and applying several laser pulses to detach them completely. All biopsied blastocysts were vitrified on day 5 or day 6 as described earlier (Van Landuyt et al., 2011). After the mutation load analysis was performed, some blastocysts with a higher mutation load (>20%; determined on the initial biopsies) were thawed for further validation of PGD procedure on multiple biopsies within the same blastocysts. Beta-hCG testing was performed 12 days after frozen thawed blastocyst transfer in a natural cycle.

2.3. DNA extraction from TE, buccal swab, blood and urine

Isolated TE cells were transferred into separately labeled 200 μl micro-centrifuge tubes (Westburg, Leusden, The Netherlands) containing 10 μl of PicoPure DNA extraction buffer with proteinase K (PicoPureDNAextraction kit, Arcturus, Mountain View, CA, USA). All samples were incubated at 65 °C for 3 h, centrifuged briefly and heated at 95 °C for 10 min to inactivate proteinase K.

Buccal swabs were collected from mother and baby in cotton plugs. The cotton swab was recovered in a sterile petri dish with the help of a sterile scalpel which was dipped into a 1.5 ml Eppendorf tube containing 1 ml HPLC water. The cells were mixed well by vortexing them for 10 s and incubated for 30 min in a thermo mixer at room temperature. The pieces of cotton swabs were removed from the solution and the cells were stored at −20 °C. The cells were pelleted by centrifuging at 14,000 rpm for 5 min and the supernatant was discarded. The cells were resuspended in 200 μl of Chelex by vortexing. The cells were incubated for 30 min at 65 °C and were re-incubated at 99 °C for 8 min. The cells were mixed by vortexing and stored at −20 °C until further use. mtDNA was collected from the supernatant after centrifuging the solution for 3 min at 14,000 rpm. Peripheral blood (10 μl) was diluted in 1 ml sterile water. Cells were spun at14,000 rpm for 5 min and the supernatant was discarded. The pellet was dried by exposing them at room temperature. Chelex (200 μl) was added to the pellet and mtDNA was extracted as described above. Similarly, 10 ml urine was collected from mother and the baby in separate tubes, which was spun at 14,000 rpm for 5 min. The supernatant was discarded and the pellet was washed twice in PBS. The cells were resuspended in 200 μl Chelex and mtDNA was extracted as described above.

2.4. PCR-RFLP in TE, buccal swab, blood and urine

PCR-RFLP procedure was performed as described elsewhere (Neupane et al., 2014; Vandewoestyne et al., 2011, 2012). Shortly, mtDNA was amplified by a 6-FAM labelled forward primer (5′-CCCA CACCCACCCAAGAACA-3′) annealing the mitochondrial genome from nt3204 to 3223 and a NED labelled backward primer (5′-TGGCCATGGG TATGTTGTTA-3′) annealing the mitochondrial genome from nt3300 to 3319 (Applied Biosystems, Foster City, CA, USA). PCR was performed in a pre-amplification reaction volume of 20 μl containing 10 μl of DNA extract and 10 μl of the master mix. The master mix consisted of 0.5 μM forward primer, 0.5 μM backward primer, 1× PCR buffer (Applied Biosystems), 10 mM dNTPmix (Applied Biosystems), 0.3 μg/μl bovine serum albumin (BSA) (Sigma Aldrich, Bornem, Belgium), 0.5 mM $MgCl_2$ (Qiagen, Germantown, MD, USA) and 1.3 U Hotstart Taq DNA polymerase (Qiagen). The conditions used to amplify the 116-bp fragment were as follows: a 15 min hot-start at 95 °C, followed by cycling at 95 °C for 1 min, 63 °C for 1 min, and 72 °C for 80 s, followed by the final elongation at 72 °C for 10 min. To achieve successful amplification TE, buccal swab, blood and urine were processed for 30, 24, 22 and 28 PCR cycles, respectively. ApaI restriction enzyme mix (1 μl) was added to the 4 μl PCR product. The restriction mix was composed of 0.01 μl ApaI

(80 U/μl, Promega Corporation, Madison, WI, USA), 0.1 μl BSA (Promega), 0.1 μl buffer A (Promega) and 0.8 μl water. Digestion was achieved by incubating specimen at 37 °C for 4 h. To ensure proper digestion of all amplicons, second digestion was achieved by adding 1 μl of ApaI restriction enzyme mix to the first digested product and incubating them for additional 4 h at 37 °C.

2.5. Capillary electrophoresis and calculation of mutation load

GeneScan 500 ROX dye Size Standard (Applied Biosystems, 401734) was mixed with formamide at the ratio of 1:24. ROX/formamide mixture (12 μl) was loaded in each well of a 96-well plate which was supplemented with 1 μl restriction digested product. The resultant mixture was loaded to ABI 3130 Genetic Analyzer and the results were analysed by GeneScan software (Applied Biosystems). After PCR, a double stranded 116-bp fragment labelled with 6-FAM as well as with NED was generated. Digestion with the restriction enzyme ApaI cleaved the mutant amplicon into a 6-FAM labelled 41-bp fragment and a NED labelled 75-bp fragment, whilst the wild-type amplicon remained undigested. After capillary electrophoresis, mutation load was calculated by dividing the peak area of the 6-FAM 41-bp fragment or the NED 75-bp fragment by the sum of this peak area and the peak area of the corresponding 116-bp fragment, multiplied by 100.

3. Results

3.1. Analysis of blastocysts

A clinical ICSI-PGD cycle was performed in a patient with m.3243A>G mutation. After ICSI in 19 MII oocytes, 15 showed normal fertilization, whereas three showed ≥3PN and one 1PN formation. All 15 zygotes underwent normal cleavage, and all of the embryos showed 7–8 blastomeres on day 3 with an average of 10% fragmentation (ranging from 0 to 35%). On day 3, eight embryos were artificially hatched, while in the rest of the embryos, an opening was made in the morning of day 5 in opposite of the developing ICM (when visible). Of the 8 embryos hatched on day 3, 6 blastocysts were obtained with variable embryo scores (Table 1). Decision was made to biopsy 3 of

Table 1
Blastocysts subjected to trophectoderm (TE) biopsy. Embryos 1, 2, 5 and 7 were warmed for analyzing multiple TE biopsies for validation. Embryo 11 was transferred back to the patient and achieved pregnancy. Mutation load in cytoplasmic fragments (CF) was similar to TE biopsies. Blastocysts were scored and graded according to Gardner's system (Gardner and Schoolcraft, 1999). For scoring: 1 indicated early blastocyst, 2 indicated blastocyst with <50% cavitation, 3 indicated expanded blastocyst, 4 indicated hatching/hatched, and 7 indicated assisted hatching/hatched blastocyst. For grading ICM: grade A indicated a tightly packed ICM with many cells; grade B, a loosely grouped ICM with many cells; and grade C, an ICM with very few cells. For grading TE: grade A indicated a TE with many cells forming a cohesive epithelium; grade B, a TE with few cells forming a loose epithelium; and grade C, a TE with very few cells (BL, blastocyst; NR, no result).

Embryo	D5 score	D5 biopsy	D6 score	D6 biopsy	Number of TE cells biopsy	Mutation load (%)	Mean mutation load ± SD
1 Warmed for validation	BL 7BC	+			2	42.8	40.3 ± 2.6
					3	42.4	
					1	NR	
					5	41.2	
					7	41.3	
					4	35.4	
					4	38.2	
					CF	40.7	
2 Warmed for validation	BL 2	−	7AA	+	1	36.5	28.2 ± 4.6
					CF	31.6	
					2	23.7	
					10	27.7	
					2	28.1	
					15	23.2	
					CF	26.6	
3	BL 2	−	7BA	+	2	0	0
					1	0	
					5	0	
4	BL 3AB	+	−	−	2	12.8	15.3 ± 3.5
					2	17.7	
5 Warmed for validation	BL 2	−	7CB	+	3	65.5	62.0 ± 4.3
					2	60.7	
					6	NR	
					6	NR	
					20	56.4	
					CF	65.2	
6	BL 7BB	+	−	−	2	0	11.4 ± 16.1
					4	22.7	
7 Warmed for validation	BL 1	−	7CC	+	3	NR	27.8 ± 4.0
					3	31.1	
					6	29.4	
					2	28.8	
					1	21.8	
					2	33.3	
					20	25.8	
					CF	24.3	
8	BL 1	−	7BC	+	2	8.8	8.4 ± 0.6
					CF	7.9	
9	BL 4AA	+	−	−	3	10.8	12.3 ± 2.1
					5	13.8	
10	BL 1	−	7 AC	+	2	7.8	11.3 ± 4.9
					CF	14.8	
11	BL 4AA	+	−	−	4	0	0 (ET)
					2	0	

these blastocysts on day 5 followed by their vitrification. The remaining embryos were further cultured until the morning of day 6 at which day 3 more blastocysts could be biopsied (Table 1). Of the 7 embryos that had not been hatched on day 3 but in the morning of day 5, two good quality blastocysts (4AA) were obtained and biopsied on day 5. The 3 others were left in culture and underwent biopsy on day 6. TE was successfully biopsied from all blastocysts (Table 1). The mutation load in the analyzed blastocysts ranged from 0% (embryo no. 3 and 11) to 65.5% (embryo no. 5). For validation, it was decided to warm the embryos detected with high level of mutation load (no. 1, 2, 5) or those of inferior quality (no.7) in order to take multiple biopsies from the entire embryo so as to verify correlation between multiple TE biopsies within blastocysts (Table 1).

3.2. Analysis of TE cells and cytoplasmic fragments

In most of the cases, multiple biopsies from each blastocyst could be taken with an average of 5 cells (ranging from 1 to 11 cells) per blastocyst recovered, except for the embryos numbered 2, 5 and 7 in which only one biopsy could be performed. From all the biopsies, including the ones after warming for validation, 4 of them (9%) failed to produce result for the PCR. Inter-TE variation analyzed within the same blastocyst showed a strong correlation of 0.90 (Pearson's correlation coefficient, r). The level of mutation among analyzed TE cells varied by 0.0–7.0%, with an exception of 22.7% in embryo 6. Anuclear cytoplasmic fragments (cf) analyzed from 5 embryos also showed strong correlation ($r = 0.95$) with the biopsied TE cells. Interestingly, the mutation load determined in the leftover cytoplasmic fragments was comparable with that in the blastocyst.

3.3. Analysis of newborn tissues

Two blastocysts were found to be mutation-free, and considering the higher quality at day 5 it was decided to transfer back embryo 11 (Fig. 1, Table 1). Transfer of this vitrified/warmed blastocyst resulted in a birth of a healthy singleton baby boy. Analysis of MELAS mutation load in tissues of the newborn baby revealed 0% mutation load in the saliva, 0% in blood cells and 0% in urine, while the mother contained 27.8%, 15.5% and 48% mutation load in these 3 tissues. Three positive controls (100% mutated mtDNA from a cybrid cell line), three negative controls (0% mutated mtDNA from a healthy volunteer) and three blanks (no template) were also analyzed to rule out technical errors.

4. Discussion

The complex mechanism of mtDNA segregation and the lack of clinical data make it difficult to define an absolute threshold that can be used when offering PGD for mtDNA mutations, although some guidelines have been reported (Hellebrekers et al., 2012). In a recent study, a threshold for MELAS of 15% was used based on data gathered in skeletal muscle and consequent clinical manifestations, including a safety margin to correct for potential errors in determining mutation load percentages. Other thresholds in current PGD practice range up to 30% for MELAS (Monnot et al., 2011). In the ideal situation, only mutation-free embryos should be eligible for embryo transfer to minimize or at least strongly reduce the risk of disease transmission. Our data demonstrate the first ever PGD case in which an mtDNA mutation-free baby, confirmed in three different tissues, is born after TE biopsy from a MELAS mutation carrier.



Fig. 1. Blastocyst scored 4AA on day 5, vitrified on day 5, mutation-free embryo and transferred (ET) in natural cycle.

Only few cases of successful pregnancies have been achieved for the MELAS mtDNA disorder so far. Specifically for MELAS, it is of utmost importance that the embryos selected for transfer preferably is mutation-free or contain a very low mutation load, due to the lack of unanimous mutation load threshold for phenotypic expression of the disease. Some individuals with identical mutation loads can manifest a different severity of this disorder, already from percentages as low as 20% (Monnot et al., 2011). In the current study, PGD analysis showed two of 11 analyzed blastocysts to be mutation-free in the proband which is encouraging for other similar patients. In addition, our data show a very high correlation between multiple TE biopsies within the same blastocysts, and also points to the possibility of using anuclear fragments to determine the mutation load. Finally, vitrification of biopsied blastocysts on day 5 or day 6 did not seem to hamper the survival rate, and led to a successful live-term after transfer of a vitrified-warmed single blastocyst.

It remains critically important to ascertain whether the detected mutation load predicted in the embryo by PGD correlates well with the mutation load in the tissues in the resulting offspring. Monnot et al. reported the birth of a healthy baby born from a MELAS patient after blastomere biopsy (Monnot et al., 2011). The transferred embryo had a mutation load of 5% in each of the two blastomeres and resulted in a baby, in whom the MELAS mutation load in the placenta and cord blood was found to be $15 \pm 5\%$ and $5 \pm 1\%$, respectively. From a mother carrying the Leigh (m.8993T>G) syndrome, a mutation-free baby was born after transfer back of two embryos that were found mutation-free using blastomere biopsy at day 3 (Sallevelt et al., 2013). Before, Thorburn et al. already reported the birth of a girl having a mutant load of approximately 4% in cord blood for a m.8993T>G mutation after PGD on blastomeres containing a low mutant load of 2.5% (Thorburn et al., 2009).

Embryo transfer in a 35% MELAS patient of a low mutation load blastocyst (12%) resulted in the birth of a boy which prevent the transmission of the mutation to the third generation as is the case in ours (Treff et al., 2012). Buccal cell mtDNA analyzed at the age of 1 month showed a mutation load of 15%. Of importance, test results performed in a commercial laboratory indicated undetectable levels of the mutation from a buccal swab and urine sediment and <10% mutation load in the blood at 5 and 12 months of age. In our study, buccal cells analyzed at the age of 3 months and blood and urine analyzed at the age of 4 months confirmed that the cells were free of MELAS mutation. The large variation in mutation load observed in different blastocysts from the same individual supports the theory that the mutant mtDNA proportion is largely determined by random genetic drift.

A second issue that remains critically important is the reliability of PGD analysis in predicting the mutation load. In patients with various mitochondrial disorders, poor correlation was found in the mutant load between first PBs and corresponding oocytes and embryos (Gigarel et al., 2011; Vandewoestyne et al., 2012) which makes that measurement of the mutation load in the PBs not a reliable method for predicting mitochondrial disorder transmission, as was supported in a recent heteroplasmic mitochondrial mouse model (Neupane et al., 2014). At the embryo or blastocyst stage, it is still unclear whether the measured mtDNA mutation load is consistent throughout the entire embryo. In mouse models, blastomeres have been shown to contain

similar levels of mtDNA heteroplasmy (Dean et al., 2003). Until now, blastomere biopsy at day 3 cleavage stage embryos has been most frequently used to diagnose mutation load in embryos. The available data until now point at relatively small differences in mutation load between blastomeres within the same embryo (Steffann et al., 2006; Treff et al., 2012) (Gigarel et al., 2011; Monnot et al., 2011), (Sallevelt et al., 2013; Vandewoestyne et al., 2012). Still, the mutation load in some embryos differed substantially between blastomeres mostly due to one or more outliers which makes single blastomere analysis perilous (Sallevelt et al., 2013; Vandewoestyne et al., 2012) and would require multiple blastomeres to be analyzed. However, multiple blastomeres biopsy on day 3 impair the subsequent embryonic developmental potential (De Vos et al., 2009). Hence, TE biopsy seems potential which also reduces the risk of mosaicism as multiple cells are analyzed simultaneously without impairing the subsequent embryonic developmental potential (Treff et al., 2012).

In order to use PGD on TE cells for mitochondrial disorders, it should be ascertained that the mutation load in TE cells is representative for the entire embryo, including the ICM cells. In heteroplasmic mice, TE was shown to be true representatives of the whole blastocyst ($r^2 = 0.90$) (Neupane et al., 2014). Similar to this mouse model, Treff et al. showed a highly concordant mutation load between the biopsied TE cells with their corresponding ICM ($r = 0.97$) in a MELAS patient, indicating the faithful representation of the entire embryo. In contrast, Lee et al. found a wider variation in levels of heteroplasmy between TE cells and ICM in non-human primates (Lee et al., 2012). Our mutation load analysis showed a very high correlation between multiple biopsies within the same blastocyst, supporting the data of Treff et al. (2012). In the current study, at least two TE biopsies were performed in most blastocysts, except #8 and #10, including the one used for embryo transfer, and that those thawed for validation. Performing two or more biopsies on a single embryo will increase the certainty of the diagnosis and help avoid mosaic error/s, if any. However, it is not always recommended though to perform multiple biopsies, which will be largely depending on the quality of the blastocyst and might affect the normal development of the embryo.

Interestingly, even the mutation level present in small cytoplasmic fragments seems to correlate very well with the biopsied TE cells, which creates an extra opportunity to assess the mutation level more accurately. Unfortunately, embryos used for multiple TE biopsies could not be used for comparing the mutation load in ICM because of the poor quality ICM in two out of four embryos (embryo #5: 7 CB and embryo #7: 7 CC). In the other embryos, attempts to isolate the ICM were not successful because of collapsing of the blastocyst which made it impossible to distinguish the ICM from surrounding TE cells. Still, it would be valuable for the future to confirm the results of Treff et al. who found comparable mutation loads between ICM and TE cells.

Next to the PGD strategy, nuclear transfer has been considered as a potential approach to avoid or reduce the transmission of mitochondrial disorders in homoplasmic mutant or individuals with higher mutation load (Craven et al., 2010; Tachibana et al., 2009; Tachibana et al., 2013). Application of these nuclear transfer techniques should be tested intensively in animal models for safety concerns and efficiency before clinical application in human should be considered. This study, demonstrating the birth of a mutation-free baby from a MELAS patient after PGD based on TE biopsy at the blastocyst level, supports the opportunity of preventing transmission of heritable mtDNA disorders in patients with mtDNA mutation disorders to reproduce healthy offspring. However, in order to ensure the safety and efficacy of the intervention, children born from affected mothers must be followed up for long-term.

Acknowledgements

We acknowledge Special Research Fund (BOF) from Ghent University who awarded PhD grant to JN (grant number 01D05611). Both Petra De Sutter and Rudy Van Coster are holders of a fundamental clinical research mandate by the Fund for Scientific Research in Flanders (FWO).

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